ENG1013 2026
System Design Document

Team F16

Team member details redacted for privacy.

Original student IDs and staff email omitted from public portfolio copy.

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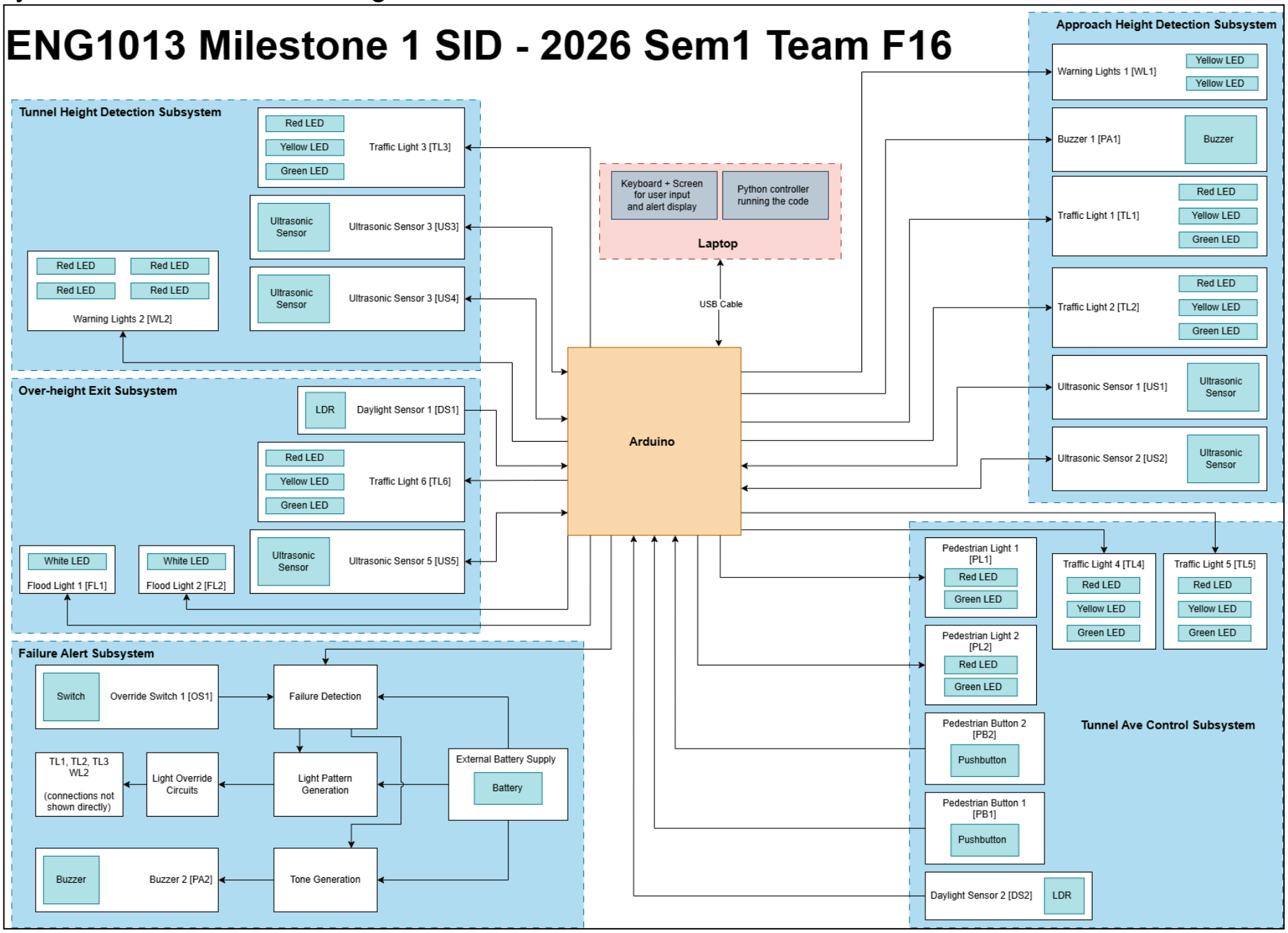
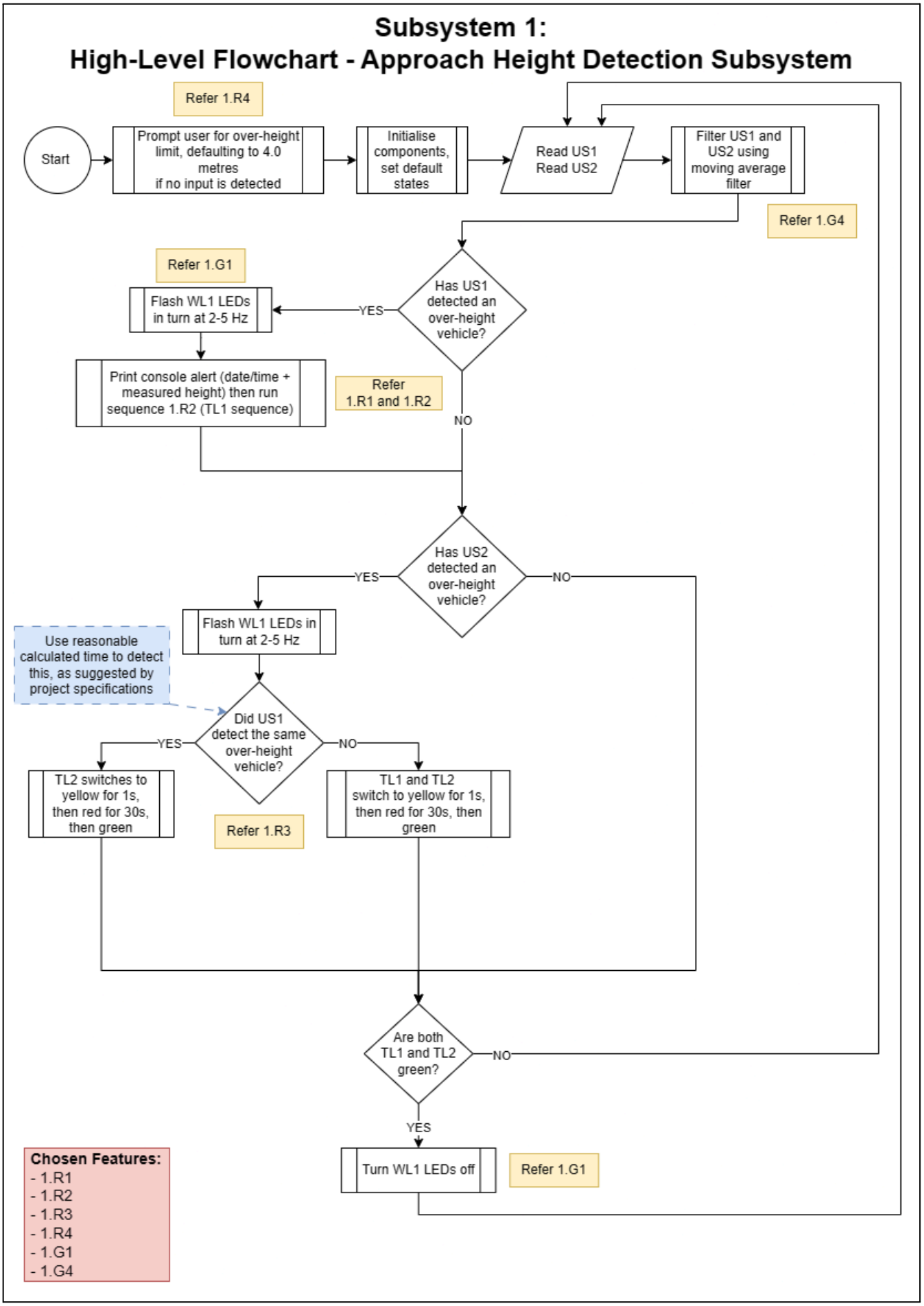
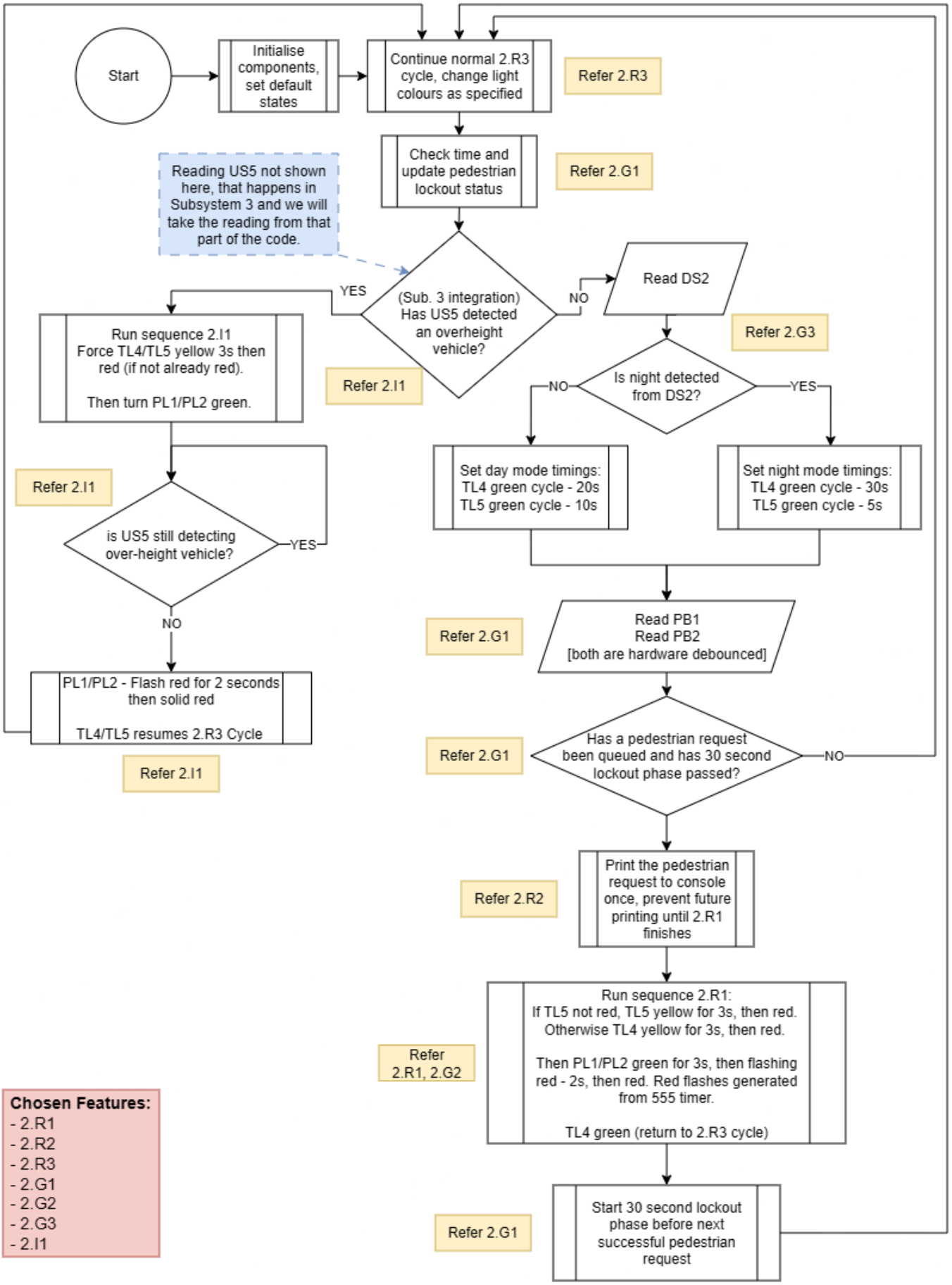
System Interaction Block Diagram:



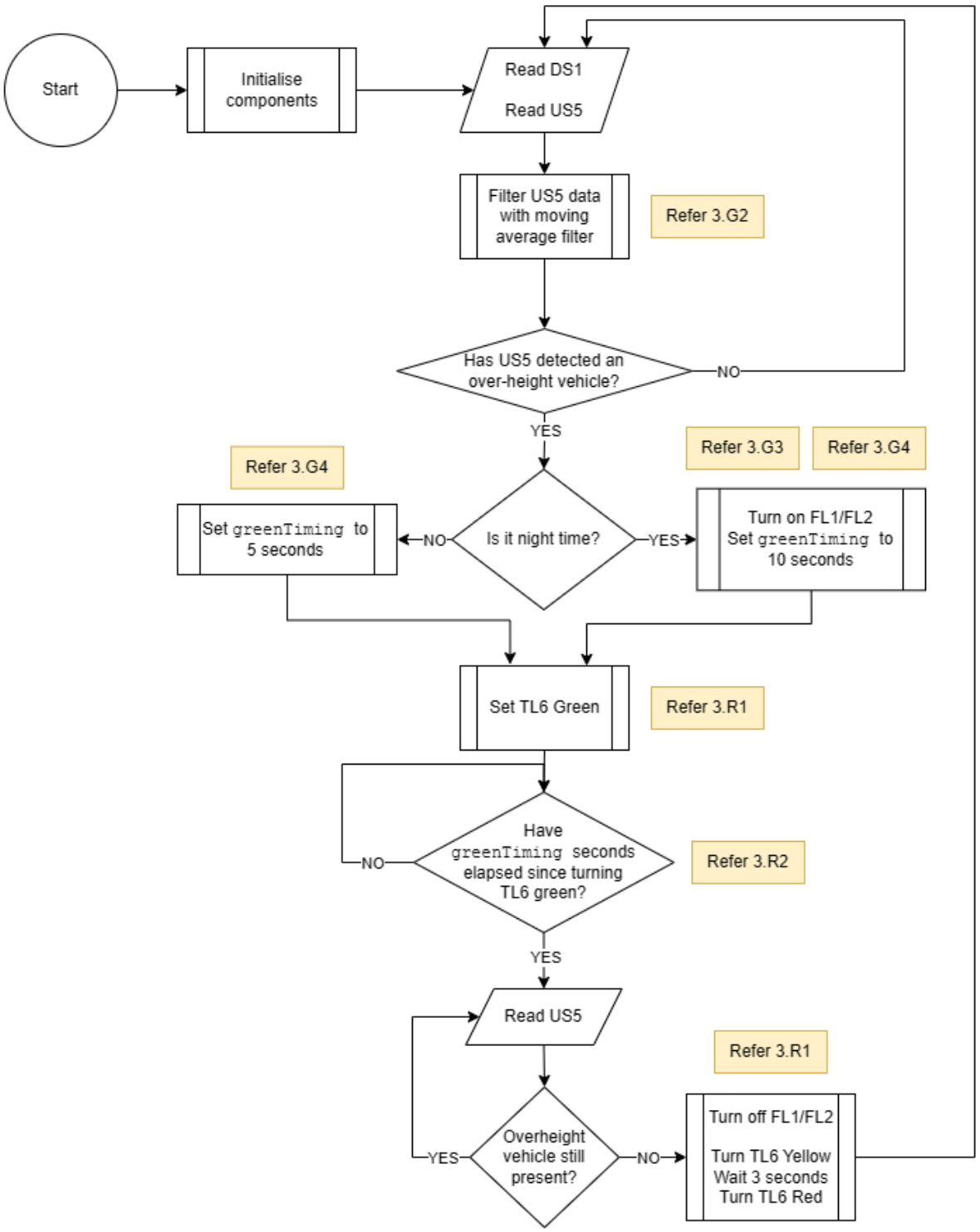
High-Level Flowcharts:



Subsystem 2:
High-Level Flowchart - Tunnel Ave Control Subsystem

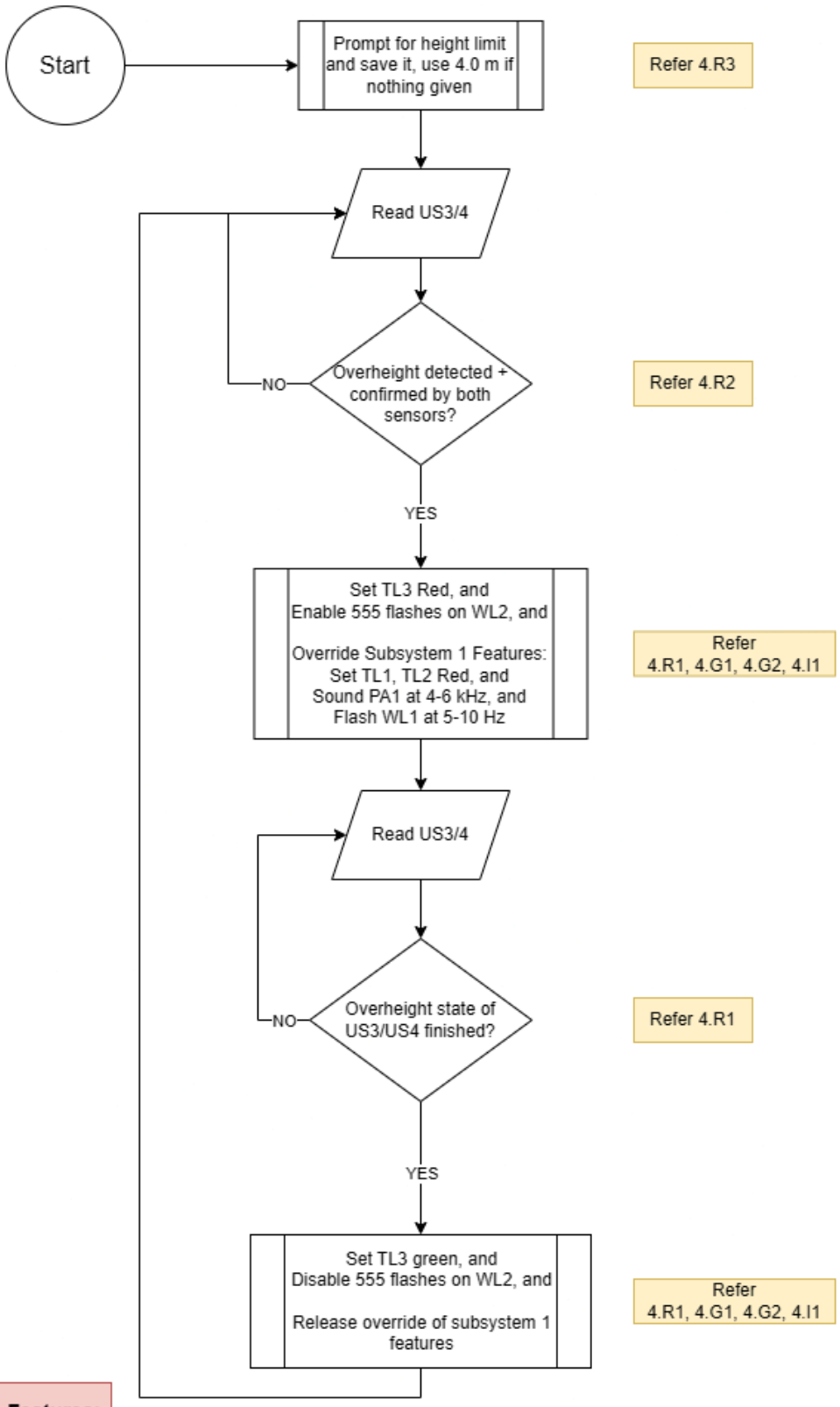


Subsystem 3:
High-Level Flowchart - Over-height Exit Subsystem



Chosen Features:
- 3.R1
- 3.R2
- 3.G2
- 3.G3
- 3.G4

Subsystem 4:
High-Level Flowchart - Tunnel Height Detection Subsystem



Chosen Features:
- 4.R1
- 4.R2
- 4.R3
- 4.G1
- 4.G2
- 4.I1

Subsystem 5 (Failure Alert Subsystem) Chosen Features:

- 5.R1
- 5.G1
- 5.G2
- 5.G3
- 5.I1

[Note: No high-level diagram is included for Subsystem 5 as it is hardware only]

Proposed Timeline:

PROJECT Milestone 1 Timeline

Group: F16

ENG1013

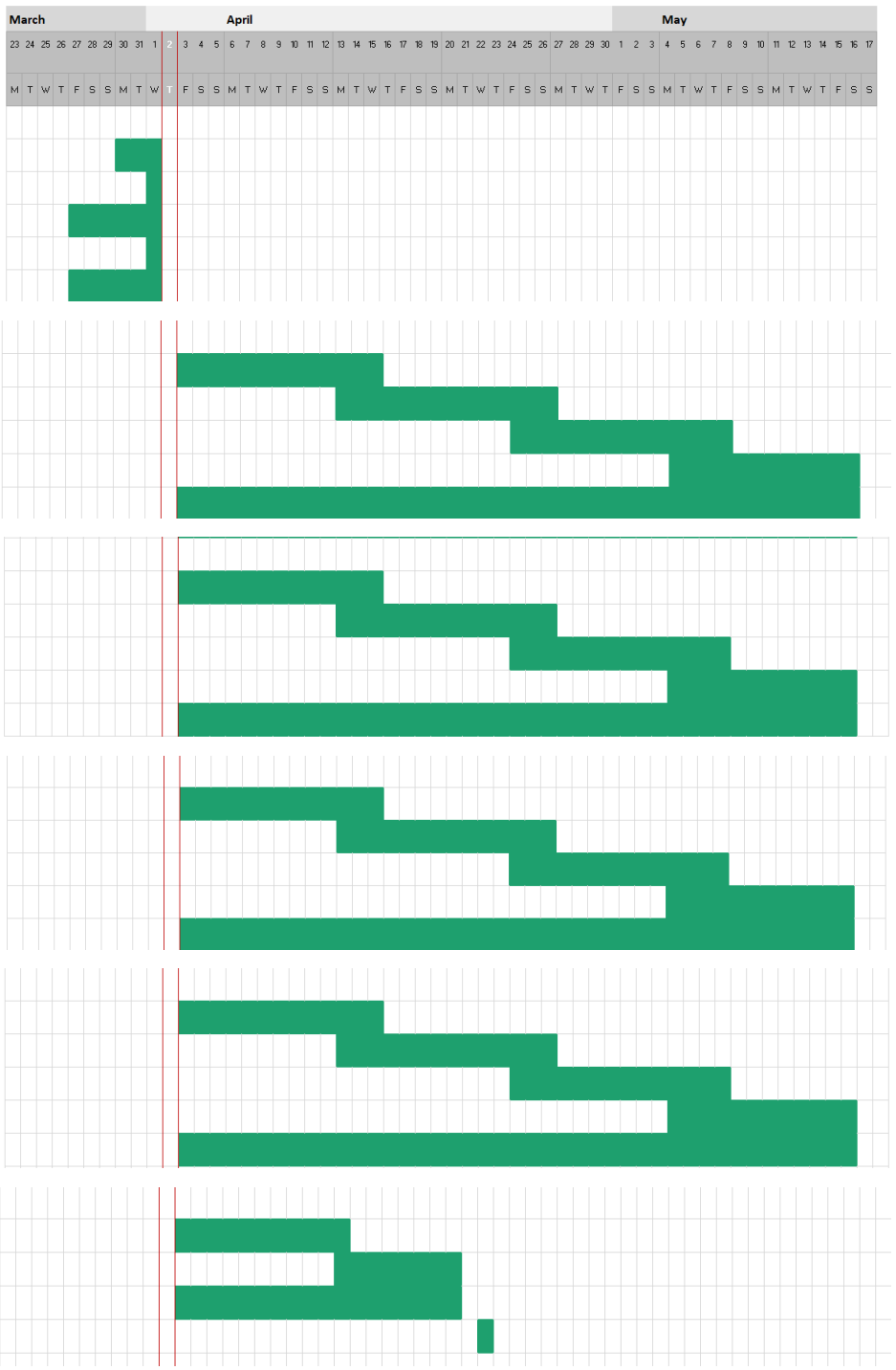
Project start date: 3/03/2026

Scrolling increment: 20

Legend: On track | Low risk | Med risk | High risk | Unassigned

Names redacted

Milestone description		Days
Milestone 1		
Gantt Chart	/26	3
System Interaction Block Diagram	26	1
High Level Flow Chart	/26	6
Communications Plan	26	1
Conflict Resolution Plan	/26	6
Subsystem 1		
Core Features Programming	5	13
General Features Programming	6	14
Core Features Wiring	6	14
General Features Wiring	5	12
Circuit Diagrams	5	43
Subsystem 2		
Core Features Programming	6	13
General Features Programming	26	14
Core Features Wiring	26	14
General Features Wiring	6	12
Circuit Diagrams	6	43
Subsystem 3		
Core Features Programming	26	13
General Features Programming	/26	14
Core Features Wiring	/26	14
General Features Wiring	26	12
Circuit Diagrams	26	43
Subsystem 4		
Core Features Programming	6	13
General Features Programming	26	14
Core Features Wiring	26	14
General Features Wiring	6	12
Circuit Diagrams	6	43
Subsystem 5		
Core Features Wiring	On	11
General Features Wiring	On	8
Circuit Diagrams	On	18
Check in with Demonstrator	On	1



Communications Plan:

Communication Methods

- Primary communication should be via the group chat on Instagram. This is where members are expected to notify the team if they are unable to attend a practical session, make arrangements regarding the project workload distribution, organise meeting times, and communicate general housekeeping and announcements.
- Secondary communications should involve more detailed information that needs to be communicated either between two members or the whole group. This should be via email in order to keep a better record without overwhelming the group chat
- Backup communication should be conducted via SMS in the event that someone is not responding via the primary or secondary communication methods.

In addition to the above virtual communication methods, the team should make an effort to meet an additional time each week outside of the allocated practical class time, particularly closer to the submission dates for each milestone. Ideally this should be conducted in person but can be conducted online via platforms such as Zoom to ensure that everyone is present and able to contribute.

- This time should be utilised to allow the group to effectively work together and communicate to receive real-time input and feedback as well as highlight questions and/or concerns.
- In addition, a ‘meeting minutes’ document should be made to keep record of the meeting dates, attendance, what was discussed, and what key decisions were made to avoid potential conflict due to disagreements. This should be a brief overview and doesn’t need a lot of detail and can be done via the Meeting Minutes template.
- Meetings should be conducted at least fortnightly. In the weeks leading up to project due dates, meetings will need to be more frequent as necessary at the discretion of the group.

Response Times

During a standard week, response times need to be within 24 hours maximum of the initial message being sent. However, responses should preferably be at a member’s earliest convenience to allow for optimal communication and organisation.

- After 24 hours has elapsed and there has been a failure to respond, a follow up message should be sent, either directly or to the group chat.
- If this occurs regularly and without valid reason, this will be escalated to a TA or team manager and taken note of for final ITP Metrics evaluation.
- In addition, if the message calls for action to be completed, this should be acknowledged with a response by the reader in adherence with the above expectations. The action itself should be completed at the member’s earliest convenience, depending on the urgency outlined by the sender.

In the week leading up to due dates and the submission of each milestone, communication should increase and responses should be within the same day, assuming the initial message is sent before 8:00pm.

- In the 2-3 days prior to the submission of a milestone, communication will need to be at its highest and responses should ideally be within 3 hours, assuming this is before the message cutoff time. However it is acknowledged that this may vary depending on whether a member is actively working on the project or not. If the message calls for the completion of an action, this should be done ASAP and prioritised above non-course-related activities.
- On the date of submission of a milestone, members need to be actively communicating and responding ASAP until the submission closes at 11:55pm OR until all of the required work and documentation has been submitted to a satisfactory standard.

Considerations

Members are expected to communicate with the rest of the group if they are unable to attend a practical class and notify everyone on the designated group chat before the start of each session.
In addition, meeting times should be organised to best accommodate everyone's needs and availability. In the event that a group member is unable to attend or cancel and make alternative arrangements (e.g. connect during an in-person meeting via Zoom), they are required to communicate their apologies and take any necessary action to ensure that they are up to date with any group decisions and discussions.

- In missing group meeting times without communicating apologies or making an effort to catch up on content discussed, members accept that they may potentially be excluded from group decisions.

Furthermore, it is acknowledged that group members may experience adverse conditions outside of their control. In the scenario that something occurs that will recurrently impact a member's ability to adhere to the above communication expectations (e.g. illness, injury, family issues etc), the member needs to communicate this to the rest of the group. If these conditions negatively impact the member's capacity to contribute to the group, the rest of the group should be notified ASAP in order for special considerations to be taken into account. Valid is required but specific details are not necessary.

- For example, if a member's ability to respond to messages is affected, the group may decide to be more lenient regarding response times and won't escalate the situation to the team manager.
- In addition, this can allow for the group to redistribute a member's work as they deem appropriate in order to assure that all due dates are met and milestones are successfully submitted on time.

However, no such adjustments and considerations can be made if the member's conditions and needs are not communicated with the rest of the group.

Conflict Resolution Plan:

Conflict Avoidance Strategies

This section will cover our team's strategies to avoid conflict, de-escalate situations and resolve disagreements such that we can ensure consistent progress, high standards of work as well as fair contribution.

Disagreements

We will have group discussions for every major decision within the project that allows each member to express their own ideas freely. Then after every member has expressed their own individual ideas, a list of advantages and disadvantages will be made that outlines each member's contributions. A group vote will be held after in order to form a decision. In the scenario that a general consensus can be made, the respective subsystem lead that the decision affects may choose to make the final call based on the decisions made.

Low-quality work

Low-quality work is defined as work that does not meet agreed project requirements, is incomplete, or contains errors that have adverse effects on the functionality of other's works. Upon identification of said errors, the team will provide feedback and set a deadline for corrections. If low-quality work keeps getting submitted by the member, temporary reassignment of tasks will be implemented and, if necessary, escalation to project managers. To prevent issues, all work will be peer-reviewed and checked against team standards before submission or integration.

Incomplete work

When incomplete work presents itself, the responsible member will be required to complete the missing assignment with a new deadline. If this occurs repeatedly, this task may be reassigned to another team member to prevent delays to the overall project. Continued failure to complete assigned tasks will be documented and reported to project manager and teaching staff.

Late submissions

All team members are expected to meet agreed deadlines. If a delay is anticipated, the member that is expecting that delay must notify the team in advance using suitable channels of communication as previously mentioned with a valid reason and revised completion time. If work is repeatedly delayed, the member responsible will see their contributions taken away from critical tasks to ensure the project is completed within the right time frame. For excessive lateness, the instances will be documented and escalated to the project manager and teaching staff as it begins to impact team performance.

Lack of participation

If a member is identified as not contributing adequately, the issue will first be addressed through communication to understand any underlying problems. If participation does not improve, responsibilities will be reassigned and tracked. Continued lack of contribution will be documented and escalated to teaching staff.

Conflict resolution plan

Step 1: Direct Communication

The issue will first be addressed privately with the individual involved, clearly outlining the concern and expected improvements.

Step 2: Team Discussion

If the issue is not resolved, it will be brought to the group to clarify expectations and agree on a solution collectively.

Step 3: Task Reallocation or Support

If necessary, tasks may be reassigned or additional support provided to ensure the project continues progressing without delays.

Step 4: Formal Escalation

If the issue persists despite prior steps, it will be formally documented and escalated to teaching staff for further action.

Meeting Minutes Template

Meeting Minutes - [Team Name Here]

Date and Time:
[Date and Time of the Meeting]

Location:
[Building and room where the meeting took place]

Meeting Attendees:
[Names of team members attending the meeting]

Apologies:
[Names of team members unable to attend the meeting and who have communicated apologies]

Absentees:
[Names of team members who did not attend the meeting and have not communicated apologies]

Facilitator:
[Name of team member who facilitated the meeting]

Minute Taker and Time Keeper:
[Name of team member who took the minutes and kept track of the time allocated/spent]

Agenda
The agenda should be agreed on before the meeting – by e-mail, group chat, etc. It should be distributed to the invitees suitably before the meeting, such that all team members can consider their contributions for each point.

AGENDA TOPIC 1			
TIME ALLOCATED		**LED BY:**	
DISCUSSION			
A reasonable summary of the discussions had during the meeting. A clear but concise detail of the issues discussed, the decisions agreed, and any reservations/opinions held by key team members.			
CONCLUSION			
The decision/s agreed upon.			

ACTION	ACTION TO BE TAKEN BY	DUE DATE
Action description (clear goal)	Person responsible for the outcome	Date to be actioned by

AGENDA TOPIC 2

TIME ALLOCATED		LED BY:	
DISCUSSION			
A reasonable summary of the discussions had during the meeting. A clear but concise detail of the issues discussed, the decisions agreed, and any reservations/opinions held by key team members.			
CONCLUSION			
The decision/s agreed upon.			

ACTION	ACTION TO BE TAKEN BY	DUE DATE
Action description (clear goal)	Person responsible for the outcome	Date to be actioned by

AGENDA TOPIC 3			
TIME ALLOCATED		LED BY:	
DISCUSSION			
A reasonable summary of the discussions had during the meeting. A clear but concise detail of the issues discussed, the decisions agreed, and any reservations/opinions held by key team members.			
CONCLUSION			
The decision/s agreed upon.			

ACTION	ACTION TO BE TAKEN BY	DUE DATE
Action description (clear goal)	Person responsible for the outcome	Date to be actioned by

AOB

[Any other business that needs discussion]

Next Meeting

[The date, time and place of the next meeting]